PRIYABRATA DAS

B.Tech. Biomedical Engineering, National Institute of Technology Rourkela

daspriyabrata748@gmail.com | (+91) 9040945548 | LinkedIn | GitHub

Education

National Institute of Technology Rourkela Aug 2023 – Present

B.Tech. in Biomedical Engineering, CGPA: 7.33/10 *Rourkela, India*

- GATE 2026 (Biomedical Engineering): AIR 251 (Top 6.7% nationwide).

DAV Public School, Bhubaneswar Bhubaneswar, India

- **CBSE Class XII (Science – PCM):** 90.8% 2023
- **CBSE Class X:** 96.8% 2021

Research Experience

Mechanical Characterization of Alginate Methacrylate-Based Biomaterial Systems May – Jul 2026

Research Intern, Institute of Life Sciences; Supervisor: Dr. Mamoni Dash *Bhubaneswar, India*

- Investigated the influence of hydrogel and scaffold mechanics on cell behaviour and tissue regeneration by developing photocrosslinkable GelMA-, DexMA-, and Alginate Methacrylate-based biomaterial systems for regenerative medicine applications.
- Combined experimental biomaterials research with computational modelling by performing mammalian cell culture, oscillatory rheological characterization, and MARTINI 3 coarse-grained molecular dynamics simulations in GROMACS to study biomaterial–membrane interactions.
- Independently designed and executed biomaterial fabrication, mechanical characterization, cell-based studies, and computational workflows throughout the internship.

Biomedical Image Analysis using Machine Learning and Deep Learning May – Jun 2025

Research Intern, Indian Institute of Science; Supervisor: Prof. Debnath Pal *Bengaluru, India*

- Investigated machine learning and deep learning approaches for biomedical image analysis using 5,403 dental radiographs, focusing on automated feature extraction, anomaly detection, clustering, and semantic segmentation.
- Developed reproducible computational pipelines using Python, HOG, LBP, PCA, K-Means, DBSCAN, Isolation Forest, VGG16, CNNs, and U-Net, and documented the implementation as an open-source GitHub repository. [GitHub]
- Compared conventional machine learning and deep learning models to evaluate their effectiveness for biomedical image analysis, strengthening expertise in computer vision and AI-driven healthcare applications.

Analysis of Mechanical Properties of Soft Gels May – Jun 2024

Research Intern, Institute of Life Sciences; Supervisor: Dr. Mamoni Dash *Bhubaneswar, India*

- Investigated hydrogel systems for tissue engineering by combining collagen and alginate hydrogel fabrication with mammalian cell culture and oscillatory rheology to study the relationship between material properties and biological performance.
- Performed RNA isolation, cDNA synthesis, RT-qPCR, IVIS imaging, and preclinical animal studies, gaining practical experience in molecular biology and translational biomaterials research.
- Assisted in the design, fabrication, and characterization of biomaterial systems while integrating molecular biology techniques with tissue engineering workflows for regenerative medicine studies.

Molecular Diagnostics and Clinical Biomedical Research Nov – Dec 2023

Research Intern, ICMR–Regional Medical Research Centre; Supervisor: Dr. Debdutta Bhattacharya *Bhubaneswar, India*

- Explored molecular diagnostics and clinical microbiology workflows for infectious disease detection through DNA extraction, GeneXpert CBNAAT, VITEK 2 Compact, CLIA, antimicrobial susceptibility testing, E-test, and Qubit fluorometry.
- Gained practical experience with automated clinical diagnostic platforms for molecular diagnostics, microbial identification, and immunological assays used in translational biomedical research.
- Developed an understanding of standardized clinical laboratory workflows, biosafety practices, quality assurance procedures, and the application of molecular diagnostics in healthcare.

Industry Experience

Clinical AI and Predictive Healthcare Analytics Jan 2026 – Present

Research Analyst, Fluxion Enterprise *Remote*

- Developed multimodal clinical AI pipelines integrating vital signs, laboratory investigations, and clinical notes for continuous ICU risk assessment using longitudinal patient data.
- Designed predictive machine learning models from longitudinal electronic health records through feature engineering, temporal data integration, missing-data handling, and rigorous model validation.

Selected Research Projects

Brain Tumor MRI Segmentation using Deep Learning with GAN Oct – Dec 2025
Research Project, NIT Rourkela; Supervisor: Prof. Sobhan Kanti Dhara

- Investigated automated brain tumor segmentation from MRI images by combining U-Net segmentation with GAN-based synthetic image generation to improve segmentation performance and model generalization.
- Developed and documented the complete deep learning workflow using PyTorch, DCGAN, and U-Net as an open-source GitHub repository. [GitHub]

Machine Learning-based Counterfeit Drug Detection Jun 2026 – Present
Short-Term Research Project, ICMR; Supervisor: Dr. Sangram Keshari Samal

- Investigating machine learning approaches for counterfeit pharmaceutical detection to support drug authentication and pharmaceutical quality assessment.
- Exploring data-driven feature extraction, classification, and predictive modelling strategies for computational drug verification.

Muscle Fatigue Monitoring using Surface Electromyography Jan – Apr 2025
Course Project, NIT Rourkela; Supervisor: Prof. Earu Banoth

- Designed a wearable biomedical instrumentation system for real-time forearm muscle fatigue monitoring using surface electromyography.
- Integrated Arduino-based hardware and software workflows for physiological signal acquisition, processing, and threshold-based fatigue detection. [GitHub]

Publications and Research Manuscripts

Priyabrata Das, "Advancing Hepatitis-B Treatment through CLIA-Based Immune Profiling and Innovative Drug Development Strategies," presented at the *Young Scientists Conference (YSC), India International Science Festival (IISF)*, DBT-THSTI Campus, Faridabad, India, Jan. 2024. DOI: 10.13140/RG.2.2.20983.64167

Priyabrata Das, "MARTINI 3-Based Molecular Dynamics Investigation of Nanoparticle–Membrane Interactions," manuscript completed; intended for submission to a peer-reviewed journal, 2026.

Priyabrata Das, "DCGAN-Based Synthetic MRI Augmentation for Brain Tumor Segmentation," manuscript completed; intended for submission to a peer-reviewed journal, 2026.

Patents / Intellectual Property

Multipurpose Cleaning Tool
Industrial Design Application No. **459636-001**, filed through NRDC / Indian Patent Office; co-developed a modular cleaning device with an adjustable telescopic mechanism and rotating cleaning assembly for elevated and difficult-to-access surfaces.

AGRI-FLOW – A Compact Vertical Dryer
Patent / Industrial Design Application filed through NRDC / Indian Patent Office; co-developed a compact agricultural dryer integrating vertical drying, automated moisture monitoring, and controlled drying for post-harvest loss reduction.

Honors & Awards

Graduate Aptitude Test in Engineering (GATE) 2026 – Biomedical Engineering	2026
All India Rank (AIR) 251, placed among the **Top 6.7%** of candidates nationwide	
Campus Innovation Challenge, IIEC, IISER Berhampur – Top 3 Finalist	2024–25
Young Scientists Conference, India International Science Festival (IISF) – Research Poster Presentation	2024
National Talent Search Examination (NTSE) – Scholar (Top 1%)	2020–21
Indian Olympiad Qualifiers in Physics (IOQP) – Top 10%	2021–22
Vidyarthi Vigyan Manthan (VVM) – State Level Representative, Odisha	2019–20
Singapore and Asian Schools Math Olympiad (SASMO) – Gold Award	2018–19

Certifications

IELTS Academic – Overall Band 7.5 (CEFR C1)	2026
Python for Machine Learning & Data Science – Jose Portilla (Udemy)	2025

Professional Development

- Attended the *INAE Technology Conclave 2026*, Indian National Academy of Engineering (INAE), Siksha 'O' Anusandhan (SoA) University, Bhubaneswar, India.

Leadership & Service

Crime Codon, NIT Rourkela – Organization Head	2024–25
TRUST: Say No To Blind-Beliefs – Volunteer	2023–Present
TEDx DAV Public School Unit-VIII – Operations Team	2021–22